Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, DECEMBER 2023

ENERGY AUDITING [ELE 4082]

REVISED CREDIT SYSTEM

Time: 3 Hours **20 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A.	List any THREE sources of electricity. Comment on their long-term availability.	***03***
1B.	A Variable Frequency Drive is to be installed for a fan. The initial investment is 4 lakh rupees and cashflow at the end of 1st, 2nd and 3rd year are 1.70 lakh, 1.72 lakh and 1.75 lakh rupees respectively. Calculate Net Present Value at 9 % discount rate and check whether this project is feasible or not?	***03***
1C.	Explain the need for appliance manufacturers to meet the standards and labelling requirements required by the Bureau of Energy Efficiency. Name any TWO mandatory appliances that are needed to meet standard and labelling obligations as per Bureau of Energy Efficiency requirements	***04***
2A.	A power station has to meet the following demand: Group A : 200 kW between 8 A.M. and 6 P.M. Group B : 100 kW between 6 A.M. and 10 A.M. Group C : 50 kW between 6 A.M. and 10 A.M. Group D : 100 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6 A.M. Determine (i) Maximum demand (ii) Total units generated per day (iii) Load factor.	***03***
2B.	Why is power transmitted at HIGH voltages? Give any TWO reasons with the help of suitable equations.	***03***
2C.	The load on installation is 800 kW, 0.8 lagging pf which works for 3000 hours per annum. The tariff is Rs 100 per kVA plus 20 paise per kWh. If the power factor is improved to 0.9 lagging by means of loss-free capacitors costing Rs 60 per kVAR, calculate the annual saving effected. Allow 10% per annum for interest and depreciation on capacitors.	***04***
3A.	Explain energy efficiency in terms of lighting.	***03***
3B.	A three-phase induction motor has the following details: Name plate details: 55 kW, 415V, 95A, 0.90 PF, 50 Hz Operating load details: 410V, 75A, 0.80 PF, 48 Hz Calculate the loading percentage and rated efficiency of the motor.	***03***
3C.	With the help of a neat sketch, explain the working of Vapor Compression Refrigeration System.	***04***

4A. A shell and tube heat exchanger are used to increase the temperature of furnace oil from a temp of 60 degC to 120 degC using steam as the heating medium. The oil flow rate is 3500 kilo-liter/hr. The density of furnace oil is 0.89Kg/liter.

Calculate the amount of steam required in tonne/hr to heat the furnace oil, if the specific heat of furnace oil is 0.5 kCal/kg-degC.

The total enthalpy of steam is 2733 kJ/kg. The condensate leaves the heat exchanger at 397 kJ/kg.

Note: 1 Kilo Calorie = 4.184 Kilojoule. ***03***

4B. Give any THREE reasons for using steam in industrial heating applications. ***03***

4C. Evaluate the option of boiler replacement for the following boiler with a new boiler of 84% efficiency. The cost of the new boiler is Rs 30.00 lakh.

Data of present boiler:
Average steam generation from the boiler: 5000 kg/h
Fuel used: furnace oil
Enthalpy gained by the steam in boiler: 600 kcal/kg of steam
Cost of furnace oil: Rs 15000 per ton (Rs. 15 per kg)
Gross calorific value of the fuel: 10000 kcal/kg
Annual operating hours of the boiler: 6000 h
Boiler efficiency: 80% ***04***

5A. Propose any SIX ways by which the energy efficiency of buildings is improved. ***03***

5B. In a building it was proposed to install a roof top solar system to meet the electrical requirements of the building. The total monthly billing units are 1000 units. The cost of tariff is Rs.6/kWh. The total roof area of the building is 1200 sq. ft including the shading area of 20%.

(i). Calculate the capacity of solar panels in kW that can be installed if 1 kW of roof top solar system requires 10 sq. mts of shadow free area. (1 sq.mt = 10.7369 sq. ft)

(ii). Calculate the annual energy generated for 365 days from the roof top solar panels if the sunshine hours are 5.5 hours/day. ***03***

5C. Recommend any FOUR strategies to be adopted by the government agencies to accelerate adoption of energy-efficient products. ***04***